



Assessment of Natural Radioactivity Levels in Brick Factories of Al-Muthanna Governorate, Iraq

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Abstract: Natural radioactive nuclides ^{238}U , ^{232}Th , and ^{40}K present in brick manufacturing facilities pose potential environmental, health, and economic concerns. This study employed gamma-ray spectroscopy using a NaI(Tl) detector to accurately determine radionuclide activity concentrations in ten samples collected from brick factories, Iraq. The investigation evaluated several critical health risk parameters, including radium equivalent activity, excess lifetime cancer risk, absorbed dose rate, and gamma representative index. The measured specific activities for ^{238}U , ^{232}Th , and ^{40}K ranged from 32.67 ± 1.22 to 34.87 ± 1.26 Bq/kg, 24.65 ± 0.92 to 38.84 ± 1.16 Bq/kg, and 405.76 ± 4.91 to 419.92 ± 5.01 Bq/kg, respectively. All calculated radiation hazard indices were found to be within the permissible limits established by international regulatory organizations as recommended by Organisation for Economic Co-operation and Development (OECD), United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR), and International Commission on Radiological Protection (ICRP). The findings indicate that natural radioactivity levels in these facilities pose no significant health risks. Specifically, both occupational workers and the surrounding population remain protected under current operational conditions. These results provide important baseline data for radiation safety assessment in the brick manufacturing industry and demonstrate compliance with international safety standards.

Keywords: Natural radioactivity; Gamma-ray spectrometry; Brick factories; Al-Muthanna Governorate

1 Introduction

Radiation is the physical process where energy is emitted as particles or electromagnetic waves. Human exposure to ionizing radiation is a continuous phenomenon. Approximately 87% of the total dose originates from natural environmental sources, while the remaining fraction is attributed to anthropogenic activities [1, 2]. These natural sources are categorized into cosmic radiation, terrestrial sources, and internal emitters. Significant contributions arise from primordial radionuclides, specifically Uranium (^{238}U), Thorium (^{232}Th), and Potassium (^{40}K) [3]. Due to their long half-lives, these radionuclides remain persistent in the environment and are integrated into the geological composition of the Earth's crust.

The brick manufacturing industry is a critical sector where monitoring natural radioactivity is essential. Bricks are produced from soil and clay, which contain varying concentrations of radioactive elements depending on their geological origins. As bricks are a primary building material, elevated radionuclide levels can pose environmental and health challenges. These levels potentially increase the radiological burden on factory workers and residents of buildings constructed from these materials [4].

This study quantifies the specific activity of natural radionuclides (^{238}U , ^{232}Th , and ^{40}K) in brick samples collected from factories within the Al-Muthanna Governorate, Iraq. Using gamma-ray spectrometry, the research evaluates key radiological hazard indices, including radium equivalent activity, absorbed dose rates, and excess lifetime cancer risk. The results are compared against safety thresholds established by international regulatory bodies, such as United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) and the International Commission on Radiological Protection (ICRP) [5].

2 Methods and Instruments

2.1 Sample Collection and Preparation

Ten samples were collected from the most productive brick factories in Al-Muthanna Governorate, Iraq. The sampling locations were identified using GPS, as shown in Table 1.

Table 1. Sampling locations and coordinates of brick factories in Al-Muthanna Governorate

Factory Code	Names of Brick Factories	Coordinates	
		Latitude (N)	Longitude (E)
M1	Haider Yahya Hamid	31.10783	45.52740
M2	Hamed Abdul Sahib Ali	31.10588	45.53507
M3	Aqeel Hussein Hashem	31.06780	45.32086
M4	Tawfiq Mutasher Ajami	31.08625	45.36280
M5	Imad Hussein Mohammed	31.10510	45.27250
M6	Youssef Jassim Latif	31.24477	45.22623
M7	Jassim Hamoud Hassoun	31.210379	45.26420
M8	Abdu Al-Rahman Mohammed Awda	31.24201	45.21637
M9	Majed Abbadi Hassoun	31.09111	45.53113
M10	Aboud Fadhl Ali	31.10817	45.52840

Samples (1 kg each) were pulverized, sieved (2 mm), dried at 110 °C, and sealed in Marinelli beakers for 30 days to reach secular equilibrium. Gamma-ray spectrometry, employing an NaI(Tl) detector, was used to quantify the specific activities of ^{238}U , ^{232}Th , and ^{40}K . To ensure accuracy, background radiation was recorded and subtracted from all measurements.

2.2 Gamma-Ray Measurement

As illustrated in Figure 1, an ORTEC gamma-ray spectrometer equipped with a 3" × 3" NaI(Tl) scintillation detector was employed to measure the radioactivity of the brick samples. To minimize background radiation interference, the detector was housed within a cylindrical lead shield. The detection system integrates a high-voltage power supply, a preamplifier, and an amplifier. These components are coupled with an ORTEC Digi Base multi-channel analyzer (MCA) featuring 1024 channels. Spectral acquisition and nuclear data analysis were performed using the MAESTRO software package.

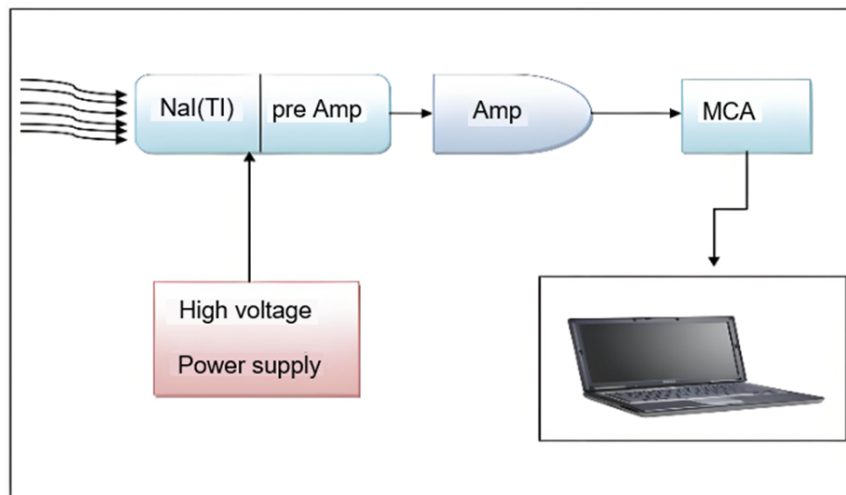


Figure 1. Diagram of the NaI(Tl) detector's apparatus setup [6]

The primary objective of energy calibration is to establish a linear relationship between the incident gamma-ray energy and the resulting pulse amplitude. This amplitude is generated by the detector crystal [7]. In this study, energy calibration was performed using a set of standard radioactive sources: ^{22}Na (511 keV and 1274.5 keV), ^{60}Co (1173.2 keV and 1332.5 keV), and ^{137}Cs (661.7 keV). Table 2 details the characteristic energy levels of each source alongside

their corresponding peak positions (channel numbers). The resulting calibration curve, which illustrates the linear correlation between gamma energy and channel number, is presented in Figure 2.

Table 2. Properties of radioactive sources utilized in this study with a comparison of experimental ($I\gamma$) [8]

Source	Energy (keV)	Channel No.	I_γ (%)	A° (μci)	Production Date
Na-22	511	201	100	1	Oct. 2009
Na-22	1274	474	99.94	1	Oct. 2009
Co-60	1173.24	450	99.97	1	Dec. 2009
Co-60	1331.23	512	99.98	1	Dec. 2009
Cs-137	661.66	263	85.1	1	Apr. 2009

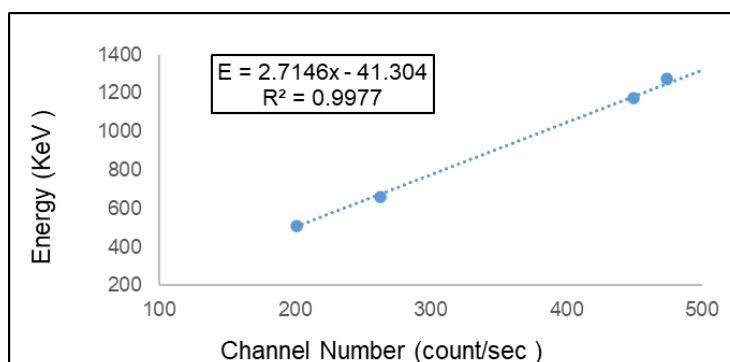


Figure 2. The energy calibration for NaI(Tl) detector used in this work

The detecting efficiency (ε) is defined as the ratio between the number of emitted pulses and the number of gamma-ray photons incident on the detector [9]:

$$\text{Efficiency}(\varepsilon) = \frac{N_{net}}{AI_{\gamma}t} \times 100\% \quad (1)$$

where, N_{net} is net area per unit time under the photopeak, A is the activity of the sample at the time of measurement (Bq), I_{γ} represents the emission probability (intensity) of gamma-ray and t is the measurement time in seconds. The activity (A) is calculated using the following decay equation [9]:

$$A = A^{\circ} e^{-\lambda t} \quad (2)$$

$$\lambda = \frac{\text{Ln } 2}{T_{1/2}} = \frac{0.693}{T_{1/2}} \quad (3)$$

where, A° is the activity at production time $t = 0$, λ is the decay constant, t is the elapsed time since manufacture, and $T_{1/2}$ is the half-life of the source. In this study, standard sources with known energies were utilized to calibrate the efficiency of the NaI(Tl) detector. The radioactivity was measured using a counting time of 200 seconds, and the resulting efficiency curve is presented in Figure 3.

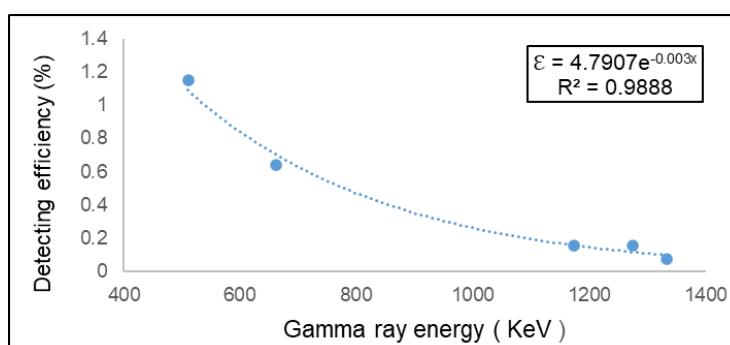


Figure 3. The efficiency calibration for NaI(Tl) detector used in this work

From Figure 3, it is found the exponential relation with correlation (0.9888) as following:

$$\varepsilon = 4.7907e^{-0.003E} \quad (4)$$

where, ε represents efficiency, E represents the energy (keV).

Figure 3 shows that the detector's efficiency peaked at 511 keV, resulting from complete incident photon absorption via the photoelectric effect. As energy increased, the efficiency gradually decreased. This reduction is attributed to the Compton effect, which causes energy loss through scattering. The photoelectric emission peaks from the ^{238}U , ^{232}Th , and ^{40}K , decay series were utilized to identify radionuclides in the samples. Detection efficiency was computed using Eq. (1) and is summarized in Table 3.

Table 3. The energy and efficiency of the radionuclide to a detector

Isotopes Daughter	Energy (keV)	Efficiency (%)
^{238}U (^{214}Bi)	1764.49	0.012
^{232}Th (^{208}Tl)	2614	0.008
^{40}K	1460.75	0.016

2.3 Theoretical Calculations

Specific activities and radiological hazard indices were calculated using standard Eqs. (5)–(11).

(1) Specific activity

The specific activity (A), of the gamma- emitting radionuclides in the samples, expressed in (Bq/kg), is calculated using the following equation [9]:

$$A = \frac{N}{\varepsilon T_c \gamma_d M_s} \quad (5)$$

where, N is the net count, ε represents the detector efficiency, T_c is the counting live time, γ_d is the gamma emission probability, M_s is the sample mass.

(2) Equivalent activity of radium (Ra_{eq})

It is used to determine the risk of a specific activity, which is given in Bq/kg. was estimated using the equation [10]:

$$Ra_{eq}(\text{Bq/kg}) = A_{Ra} + (1.43)A_{Th} + (0.077)A_K \quad (6)$$

where, A_{Ra} , A_{Th} and A_K represent the corresponding activity concentrations of (^{226}Ra , ^{232}Th , and ^{40}K) in (Bq/kg).

(3) Absorbed dose rate (D_r)

The absorbed dose rate in air (D_r), at one meter above the ground surface, resulting from the radioactivity concentrations of ^{238}U , ^{232}Th , and ^{40}K , is determined using the following equation [11]:

$$D_r(\text{nGy/h}) = (0.462)A_{Ra} + (0.622)A_{Th} + (0.041)A_K \quad (7)$$

where, the numerical coefficients (0.462), (0.622), (0.041), represent the dose conversion factors for these naturally occurring radionuclides.

(4) Hazard indices (H_{ex} , H_{in})

External exposure (H_{ext}) is utilized to assess the biological risk of radionuclides that emit naturally occurring gamma rays. This index is calculated using the following equation [11]:

$$H_{ext} = \frac{A_{Ra}}{370} + \frac{A_{Th}}{250} + \frac{A_K}{4810} \quad (8)$$

Internal exposure (H_{int}) refers to the inhalation of radon gas and its short-lived progeny. This index is calculated using the following equation [11]:

$$H_{int} = \frac{A_{Ra}}{185} + \frac{A_{Th}}{259} + \frac{A_K}{4810_{int}} \quad (9)$$

(5) Gamma representative index (I_γ)

Radiation risks associated with specific radionuclides of ^{238}U (226Ra), ^{232}Th , and ^{40}K . were evaluated using the following equation [11]:

$$I_\gamma = \frac{A_{Ra}}{150} + \frac{A_{Th}}{100} + \frac{A_K}{4500} \leq 1 \quad (10)$$

(6) Excess lifetime cancer risk

$$\text{ELCR} = \text{AEDE} \cdot \text{DL} \cdot \text{RF} \quad (11)$$

where, RF denotes the fatal cancer risk factor per Sievert (ICRP uses RF as 0.05 Sv^{-1} for stochastic impacts); DL represents for average lifespan (estimated to be 71.4 years); and AEDE denotes annual effective dose equivalent; and ELCR stands for excess lifetime cancer risk [12].

2.4 Background Radiation Measurement

Natural radioactivity from terrestrial materials, cosmic radiation, and system components contributes to background radiation signals recorded by measurement instrumentation. This background level varies spatially and depends on the detector's dimensions, type, and shielding configuration. The shield utilized in this study was constructed from low-background lead to ensure the absence of interfering isotopes, such as ^{210}Pb . Internally, the shield was lined with a graded-Z liner consisting of a 101.6 mm outer lead layer, followed by a 1 mm tin layer to absorb lead-induced X-rays. A final 1.5 mm copper layer was added to attenuate X-rays emitted by the tin. Copper was selected due to its low characteristic X-ray energy (8 keV), which does not interfere with the target measurement range.

Following the establishment of secular equilibrium between ^{226}Ra and ^{222}Rn , brick samples were placed in one-liter Marinelli beakers. To obtain the gamma-ray spectra, each Marinelli beaker was positioned on the NaI(Tl) detector for five hours. Background levels were recorded and subtracted from each sample's spectrum to determine the net radioactivity [12]. The resulting spectra were processed as illustrated in Figures 4 and 5.

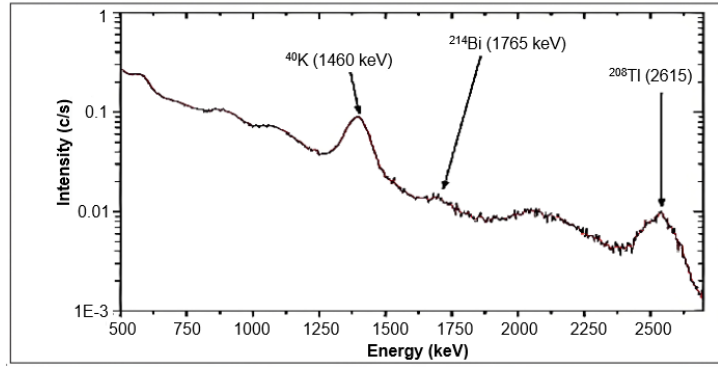


Figure 4. Gamma-ray spectrum of a typical sample measured in brick samples

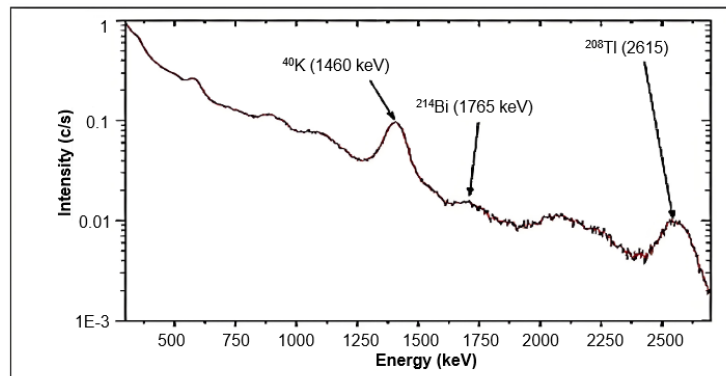


Figure 5. Gamma-ray spectrum of a typical sample measured after subtracting the background radiation in the displayed brick samples

3 Result and Discussion

3.1 The Specific Activities

Table 4 presents the specific activities of ^{238}U , ^{232}Th , and ^{40}K , measured in the brick samples. The derived radiological hazard indices—including radium equivalent activity (Ra_{eq}), absorbed dose rate (D_r), external and

internal hazard indices (H_{ex} , H_{int}), gamma representative index (I_γ), annual effective dose equivalent, and excess lifetime cancer risk—were compared against established international safety standards.

Table 4. The measured specific activities of ^{238}U , ^{232}Th , and ^{40}K in the brick samples are presented

No.	Sample Code	Specific Activity (Bq/kg) $\pm$ S.D		
		^{238}U	^{232}Th	^{40}K
1	M1	34.87 ± 1.26	32.19 ± 1.06	417.44 ± 4.98
2	M2	34.41 ± 1.25	24.65 ± 0.92	418.27 ± 4.99
3	M3	34.69 ± 1.25	30.79 ± 1.03	416.13 ± 4.97
4	M4	33.04 ± 1.22	36.19 ± 1.12	419.22 ± 4.99
5	M5	32.95 ± 1.22	32.61 ± 1.06	415.35 ± 4.97
6	M6	34.46 ± 1.25	33.95 ± 1.09	412.31 ± 4.95
7	M7	32.67 ± 1.22	38.84 ± 1.16	419.92 ± 5.01
8	M8	34.73 ± 1.26	36.68 ± 1.13	413.21 ± 4.96
9	M9	32.86 ± 1.22	37.35 ± 1.14	405.76 ± 4.91
10	M10	33.04 ± 1.22	35.59 ± 1.11	418.03 ± 4.98
	Maximum	34.87 ± 1.26	38.84 ± 1.16	419.92 ± 5.01
	Minimum	32.67 ± 1.22	24.65 ± 0.92	405.76 ± 4.91
	Average $\pm$ S.D	33.776 ± 1.116	33.848 ± 1.042	411.402 ± 2.224
	Worldwide average [11]	35	45	420

In brick factories in the Al-Muthanna Governorates of Iraq, the findings of the specific activity for ^{238}U , ^{232}Th , and ^{40}K , radionuclides are shown in Table 4. The specific activity for ^{238}U in this area ranged from 32.67 ± 1.22 to 34.87 ± 1.26 (Bq/kg), with an average of 33.776 ± 1.116 (Bq/kg). In addition, the specific activity in ^{232}Th it ranged from 24.65 ± 0.92 to 38.84 ± 1.16 (Bq/kg) with an average of 33.848 ± 1.042 (Bq/kg), whereas in ^{40}K varied from 405.76 ± 4.91 to 419.92 ± 5.01 (Bq/kg) with an average value of 411.402 ± 2.224 (Bq/kg). According to Table 4, all of the specific activities' values (^{238}U , ^{232}Th , and ^{40}K) were below the global average activity that UNSCEAR 2008 advised [4, 11]. The particular activity for the radionuclides ^{238}U , ^{232}Th , and ^{40}K in the study's current samples is displayed in Figure 6.

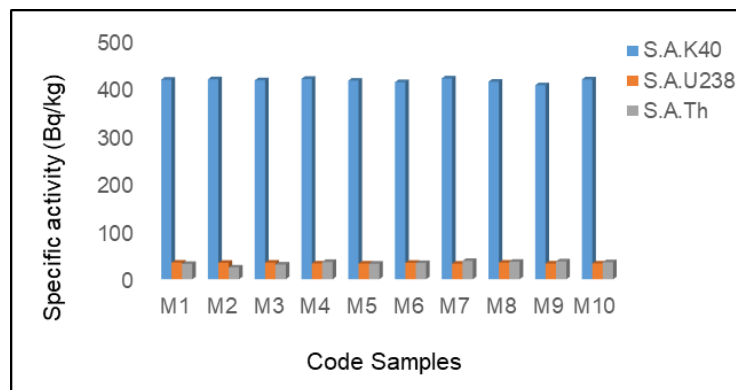


Figure 6. The specific activities of ^{238}U , ^{232}Th , and ^{40}K in the brick samples are presented

Figure 6 shows the relationship between the specific activity and sample cod for (^{238}U , ^{232}Th and ^{40}K). The variation in the radioactivity of radioactive isotopes in brick factories in Muthanna Governorate/Iraq is the result. The variation in natural radioactive nuclides in brick factories in Al-Muthanna Governorate is influenced by various geological and industrial factors. These include the quality of raw materials, the geological origin of the soil, differences in production techniques, human activities and industrial pollution, and variation in soil extraction depth. Clay or local soil content, as well as the presence of ancient sedimentary deposits, can affect the concentration of radioactive elements in the final product.

3.2 Radiological Effects

Tables 5 and 6 show the radiological effects of the brick industry in al-Muthana who are governors in Iraq (R_{eq} , D_r , H_{ex} , H_{in} , I_γ , AEDE, and ELCR). Table 5 shows radium equivalent activity found in brick samples based on the above calculation. Radium equivalent activity of brick samples varies from (101.8 to 119.984) Bq/kg, with an

average value (114.179 ± 10.685) Bq/kg. The radium equivalent activity values for each brick samples were studied and determined to be within the acceptable limits at (370 Bq/kg) [12, 13]. The determined absorbed dose rate ranged from (48.234 to 54.827) nGy/h, with an average value (53.378 ± 7.306) nGy/h. Depending on the UNSCEAR (2000), the calculated absorbed dose rate varied from roughly 45% of the global average outdoor exposure from terrestrial gamma-rays (55 nGy/h) [10, 11]. The external hazard index was calculated from (0.257 to 0.324), with an average value of the (0.308 ± 0.555), the radiation protection report said that the computed average values were less than unity [13].

Table 5. The radium equivalent (Ra_{eq}), the absorbed dose rate (D_r), external (H_{ex}), internal (H_{in}) hazard index and radioactivity level index (I_γ) in the brick samples are presented

No.	Sample Code	Ra_{eq} (Bq/kg)	D_r (nGy/h)	H_{ex}	H_{in}	I_γ
1	M1	113.058	52.965	0.305	0.399	0.416
2	M2	101.882	48.234	0.275	0.368	0.377
3	M3	110.768	51.979	0.299	0.392	0.408
4	M4	117.085	54.610	0.316	0.405	0.430
5	M5	111.579	52.246	0.301	0.390	0.411
6	M6	114.760	53.621	0.309	0.403	0.422
7	M7	119.984	54.827	0.324	0.415	0.441
8	M8	119.015	54.437	0.321	0.412	0.436
9	M9	117.521	54.664	0.317	0.406	0.431
10	M10	116.140	54.200	0.313	0.402	0.427
	Maximum	119.984	54.827	0.324	0.415	0.441
	Minimum	101.882	48.234	0.275	0.368	0.377
	Average	114.179 ± 10.685	53.378 ± 7.306	0.308 ± 0.555	0.349 ± 0.632	0.459 ± 4.583
	Worldwide average [13]	<370	<55	<1	<1	<1

Table 6. The outdoors, the indoors, the total annual effective dose equivalent (AEDE), and the excess lifetime cancer risk (ELCR) in the brick samples are presented

No.	Sample Code	AEDE (mSv/y)			ELCR $\times 10^{-3}$
		Indoor	Outdoor	Total	
1	M1	0.259	0.064	0.259	0.909
2	M2	0.236	0.059	0.236	0.828
3	M3	0.254	0.063	0.254	0.892
4	M4	0.267	0.061	0.267	0.937
5	M5	0.256	0.064	0.256	0.897
6	M6	0.263	0.063	0.263	0.920
7	M7	0.273	0.068	0.273	0.958
8	M8	0.271	0.067	0.271	0.951
9	M9	0.268	0.065	0.268	0.938
10	M10	0.265	0.066	0.265	0.930
	Maximum	0.273	0.068	0.273	0.958
	Minimum	0.236	0.059	0.236	0.828
	Average	0.262 ± 0.511	0.065 ± 0.255	0.261 ± 0.633	1.173 ± 1.083
	Worldwide average [13]	0.42	0.08	0.50	–

The internal exposure ranged between (0.368 to 0.415) with an average value (0.349 ± 0.632), therefore the calculated average values were less than unity according to the radiation protection report [13]. The calculated I_γ values for the samples of this location are presented in Table 5, the values range from (0.377 to 0.441), with an average of (0.459 ± 4.583). These calculated values are less than the international values $I_\gamma < 1$ [2, 13]. The calculated values for the average effective dose rate in indoor, outdoor, and total settings are listed in Table 6. These average values (0.262 ± 0.511 , 0.065 ± 0.255 , and 0.261 ± 0.633) mSv/y, respectively. The corresponding global values are 0.42, 0.08, and 0.50 mSv/y [4, 13]. Table 6 displays the estimated lifetime cancer risk for this samples. The range value from (0.958×10^{-3} to 0.821×10^{-3}), with average values ($1.173 \pm 1.083 \times 10^{-3}$). These findings indicate that the chance of developing cancer is around 0.117%. Figures 7, 8 and 9 show the radiological impacts (Ra_{eq} , hazard index (H_{ex} , H_{in}), and annual effective dose equivalent (in, out, and tot)) .

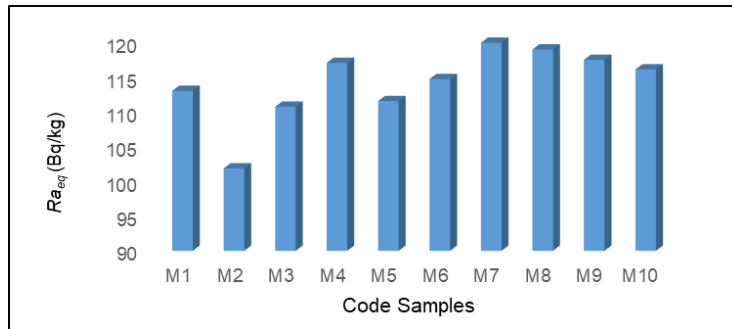


Figure 7. The radium equivalent (Ra_{eq}), in the brick samples are presented

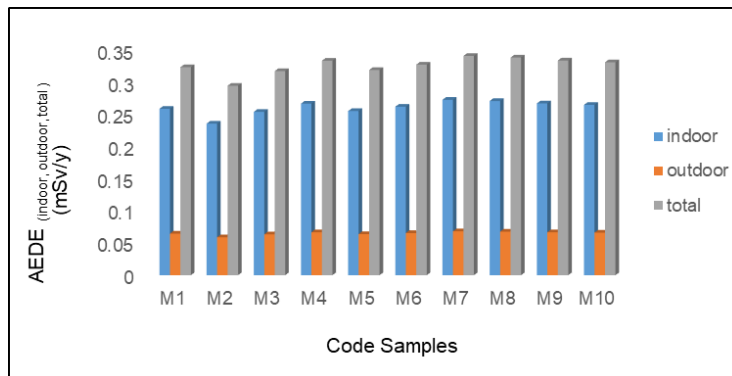


Figure 8. The outdoors, the indoor and the total annual effective dose equivalent (AEDE) in the brick samples are presented

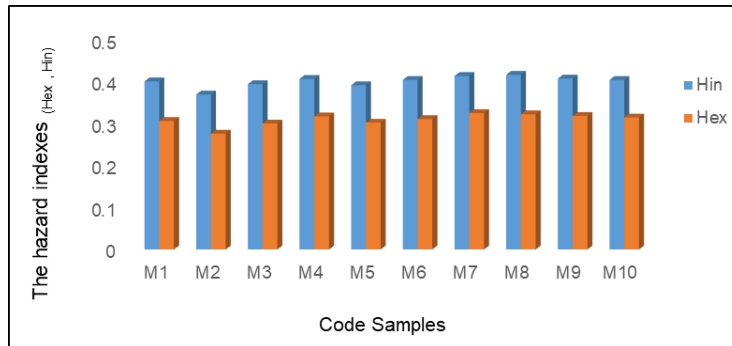


Figure 9. The hazard indexes (H_{ex} and H_{in}), in the brick samples are presented

In Figures 7, 8, and 9, the radiological effects in Al-Muthanna brick factories are influenced by three key factors: geological sources affecting natural isotope concentrations, combustion efficiency related to furnace temperatures impacting radioactive isotope concentrations, and the physical properties of bricks, including porosity and density, which determine radon exhalation rates and radiation doses.

4 Compare Gamma Emitter Results to Other Research

As indicated in Tables 7 and 8, the mean specific activities of ^{238}U , ^{232}Th , and ^{40}K obtained in this study were compared with global and local data. According to Table 7, ^{238}U activity in the current samples was lower than levels reported in Jordan, yet surpassed those recorded in Saudi Arabia, Turkey, and Iran. Similarly, the average ^{232}Th concentrations were higher than values from Saudi Arabia, Turkey, and Iran, but remained below those found in Jordan. For ^{40}K , the activities exceeded levels in Jordan and Turkey while remaining lower than those in Saudi Arabia and Iran.

Regional comparisons within Iraq, as shown in Table 7, revealed that the average specific activities of ^{238}U , and ^{232}Th , in the Al-Muthanna Governorate were higher than those in Babylon, Najaf, Missan, and Karbala. Furthermore, ^{40}K , levels were higher than in Karbala, Babylon, and Najaf, but lower than the activities recorded in Missan.

Table 7. Comparing the current study's findings with those of other nations

No.	Country	Specific Activity (Bq/kg)			Reference
		^{238}U	^{232}Th	^{40}K	
1	Saudi Arabia	14.22	14	968.19	[14]
2	Jordan	49	70	291	[15]
3	Iran	23	31	453	[16]
4	Turkey	7.4	9.5	35.7	[17]
	Iraq (Al-Muthanna)	33.776	33.848	411.402	Present study

Table 8. Comparison of the results of the current study with different sites in Iraq

No.	Governorate	Specific Activity (Bq/kg)			Reference
		^{238}U	^{232}Th	^{40}K	
1	Missan	21.19	9.72	453.91	[18]
2	Karbala	19.45	24.47	245.10	[19]
3	Babylon	16.07	9.60	271.42	[20]
4	Najaf	17.48	8.59	298.31	[21]
	Iraq (Al-Muthanna)	33.776	33.848	411.402	Present study

5 Conclusions

The findings of this study demonstrate that the specific activities of ^{238}U , ^{232}Th , and ^{40}K , in Al-Muthanna brick samples are significantly lower than the global average levels reported by UNSCEAR. These low concentrations are primarily attributed to the geological characteristics of the raw clay materials sourced from the Al-Muthanna region. Furthermore, the calculated radiological hazard indices, including Ra_{eq} and ELCR, remain well below unity and international safety limits. Consequently, the usage of these bricks in construction does not pose a deterministic or stochastic health risk to the public. Compared to previous studies in neighboring regions, these bricks exhibit a higher radiological safety profile. To maintain these safety standards, periodic screening of raw materials is essential to account for potential geochemical variations in clay deposits.

Author Contributions

Conceptualization, A.A.F.A.; methodology, A.A.F.A.; field sampling, A.A.F.A.; experimental laboratory measurements, A.A.F.A.; formal analysis, A.A.F.A.; writing—original draft, A.A.F.A.; resources, K.H.H.A.; supervision, K.H.H.A.; validation, K.H.H.A.; writing—review & editing, K.H.H.A.; data curation, R.M.O.A.; validation of formal analysis, R.M.O.A.; final review and formatting, R.M.O.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare that they have no conflicts of interest.

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